

# Research Review on Iconography

Harshkumar Beladiya(23BCE030)  
Baldaniya Yagnik (23BCE027)

Department of Computer Science and Engineering  
Institute of Technology, Nirma University

Review-2

# Introduction

- Iconography is systematic study of symbols, images, motifs and visual representation.
- Iconography examine how visual elements carried out specific meaning, beliefs and information.
- Meaning is interpreted based on the historical, cultural, religious, region ,and social contexts.
- Iconography includes interpretation of visual symbols in artwork,manuscript, religious images, mythological figures
- It also include study of modern visual symbols such as logo, icons ,emojis,etc..
- In computer science, iconography can be analyzed using computer vision and AI techniques.

# Problem Statement

- Analysis and collection of large Iconography in Manually is difficult and time consuming.
- Symbolic meaning is very complex and meaning varies on cultural, religious, and region context.
- Lack of standardized, annotated dataset limits AI-based research.
- Research is scattered across multiple disciplines.
- High variability in visual representation. The same symbol or concept may appear in different forms, styles, making consistent recognition difficult.
- Lack of labeled and annotated data

# Research Objectives

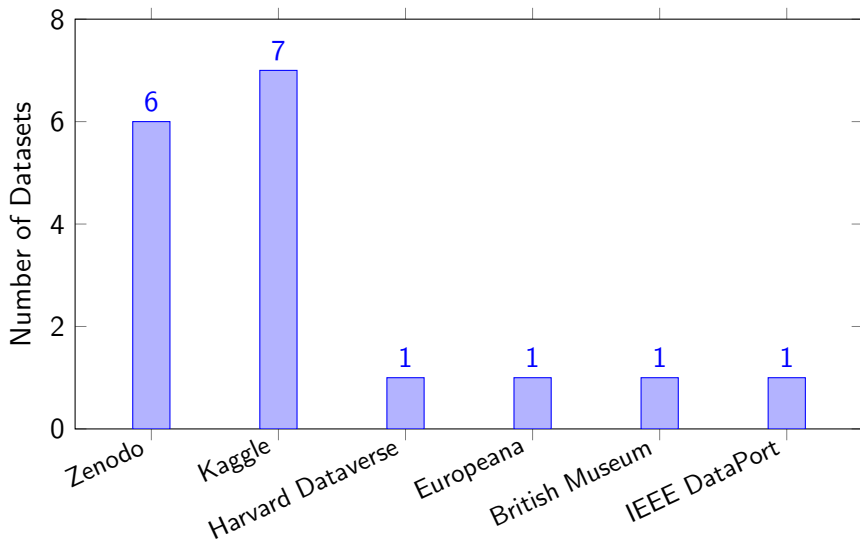
- Identify relevant datasets from multiple domains for iconography.
- Collect and analyze research papers on Iconography Concepts.
- Classify implementation paper of Iconography based on methods/sub-methods and applications.
- Making a structured taxonomy of Methods and applications.
- Identify research gaps.

# Work Progress Overview

- We explored different source for finding Dataset on Iconography in **Cultural and heritage**.
- Created Excel for various Dataset from multiple Sources
- Collected **76 research papers** from multiple Sources.
- Created structured Excel with classification.
- Identified methods and sub-methods used in Iconography in **Cultural and heritage**.
- Taxonomy refined based on collected 76 Papers.

- Sources explored:
  - Kaggle
  - Zenodo
  - Europeana
  - Museum repositories
  - Research datasets
- We Focus on datasets that is related to art, heritage, and symbolic imagery.

# Dataset Source Distribution



# Dataset Observations

- We observed that Datasets are domain-specific like one dataset only containing either paintings, or murals, or manuscripts.
- There is Limited large-scale benchmark datasets.
- Many datasets are small or specialized.
- Lack of unified dataset for iconography research.



- Till now we have collected total **76** papers.
- Sources:
  - IEEE, Springer, Elsevier, MDPI , ACM
- We selected Papers for Literature review based on:
  - Relevance to iconography
  - AI/computer vision methods on Iconography
  - Contribution to Iconography
  - Supporting paper to Iconography

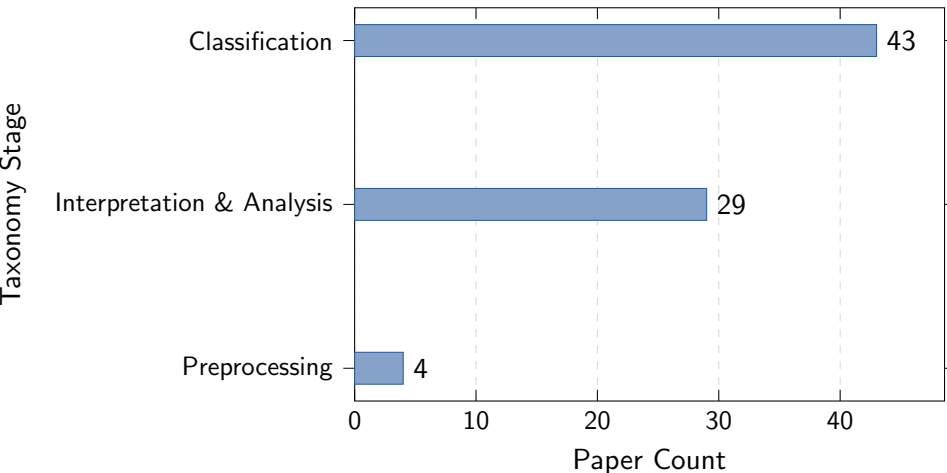
# Paper Classification

- Papers mainly grouped into three groups:
  - Core Iconography
  - AI + Iconography
  - Supporting Papers
- this classification helps to organize research and avoid overlap.

# Method-wise Distribution

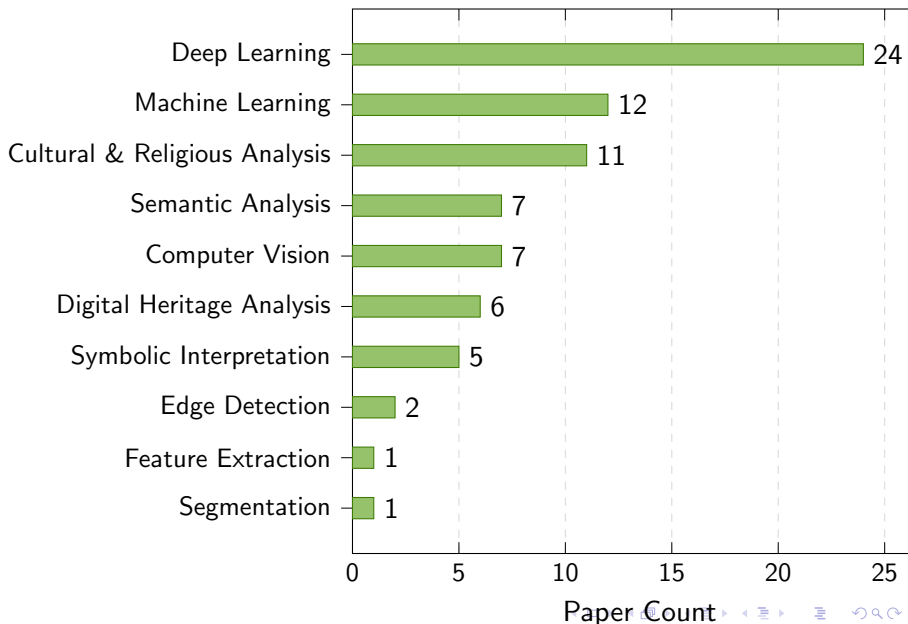
- Classification Methods:
  - Computer Vision, Deep Learning, Machine Learning.
- Interpretation Analysis:
  - Cultural Religious Analysis, Digital Heritage Analysis, Semantic Analysis, Symbolic Interpretation
- Preprocessing Methods:
  - Edge Detection , Feature Extraction , Segmentation.

# Paper Distribution by Taxonomy Stage

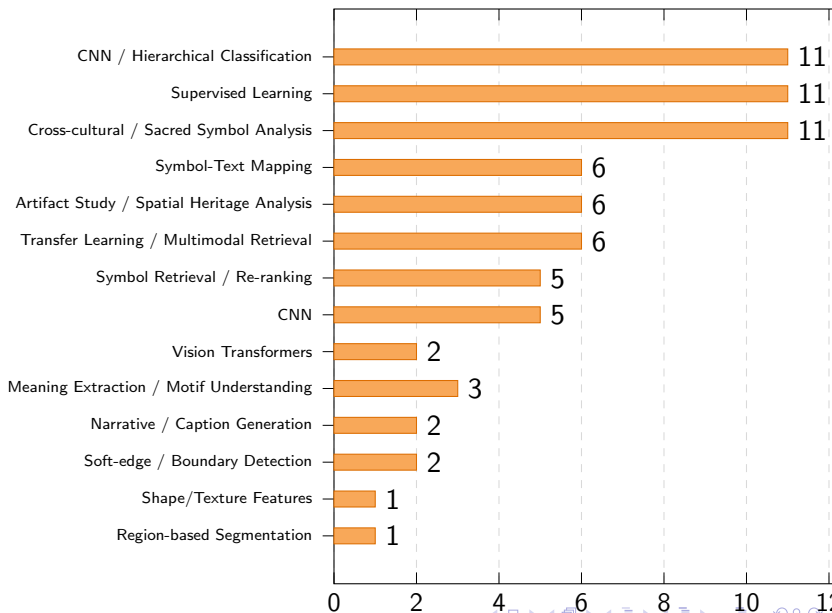


# Paper Distribution by Method

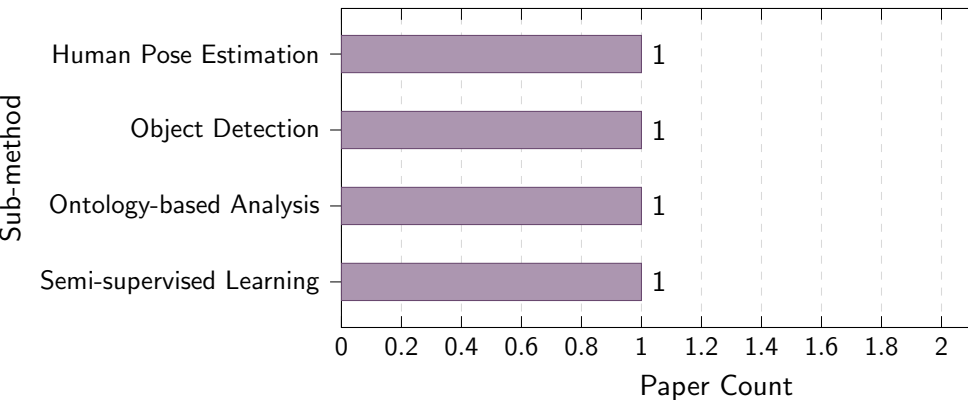
Method



# Paper Distribution by Sub-method (1/2)



# Paper Distribution by Sub-method (2/2)



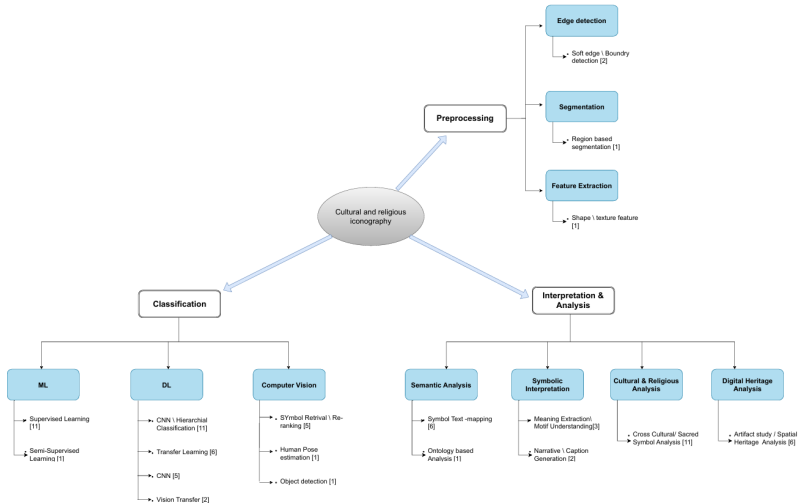


Figure: Taxonomy for mwthods



# Key Observations

- Majority papers focus on classification and retrieval.
- There has been limited work done on symbolic meaning extraction.
- Dataset availability on cultural heritage side is a major challenge.
- There is growing use of deep learning and multimodal AI in Iconography.

- Lack of standard datasets particular on Cultural heritage side.
- Limited symbolic interpretation methods on iconography.
- Weak integration of visual and textual data.
- No unified evaluation benchmarks.

- Refine taxonomy using 76 papers.
- Select specific problem for implementation on Cultural and heritage side.
- Choose suitable dataset to train a model.
- Apply AI techniques for iconography analysis.

# Conclusion

- Dataset exploration completed.
- 76 papers collected and analyzed.
- Literature structured into meaningful categories.
- Foundation prepared for next research stage.

# Thank You